

Ankit Kumar

Machine Learning Engineer | Agentic AI, RAG & Production ML Systems

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Seattle, WA | F-1 STEM OPT through May 2028

MLE with **4+ years** of experience building and **deploying ML and GenAI systems**—from hybrid RAG pipelines with retrieval, reranking, and evidence-gated generation to physics-constrained models running on embedded hardware. Takes **extreme ownership** from ambiguous requirements through implementation and validation, **challenges assumptions with data**, and brings technical and business teams together to **deliver reliable, user-focused solutions**.

EXPERIENCE

Milwaukee Electric Tool Corporation

July 2025 - July 2026

Machine Learning Engineer - Front End Innovation (FEI)

- Led end-to-end development of a physics-informed PyTorch thermal warning system combining a three-body ODE with electrical heat generation; translated requirements from Product Management into model behavior, warning logic, validation criteria, and embedded C specifications, achieving 2.2°C maximum RMSE across 30 held-out profiles and warnings up to 60 seconds before hard shutoffs.
- Cut experiment iteration from 2 hours to 40 minutes by building Azure-based ETL and training pipelines that synchronized lab and field-collected tool-usage data, automated preprocessing and model execution, and used MLflow to track datasets, parameters, metrics, artifacts, and model versions — enabling faster model iteration ahead of milestone deadlines.
- Reduced validation time from 30 days to 3 days by proving equivalent internal heat generation between matched field-use cutting and laboratory load-brake cycles; used ablation and counterfactual analyses from -20°C to 40°C to identify a sensor-induced 4–5°C error, supporting its removal and \$100K in projected hardware savings.
- Drove the PyTorch-to-embedded-C implementation in partnership with the principal firmware engineer, building a time-causal Python reference twin and Azure-hosted Streamlit app for real-time telemetry comparison and session replay; identified two release-blocking OCV/DCIR and initialization defects before they could corrupt field feedback and drive decisions off invalid data, avoiding an estimated 30+ days of firmware and validation rework.

Mitra | Mason Innovation Exchange, GMU

June 2024 - May 2025

Founder & Lead ML Engineer - Selected University Venture

- Owned development of an AI-powered university decision-support platform from problem definition and architecture through evaluation, deployment, and a controlled alpha with 200+ prospective student users, translated ambiguous information needs into requirements for ingestion, retrieval, agent behavior, and evidence-grounded responses while directing student contributors across infrastructure and application development.
- Led the design and implementation of an event-driven AWS data platform (Scrapy, Lambda, SQS, S3, Glue, Athena, PySpark) to process 8K+ university webpages per crawl, combined deterministic rules with GPT-3.5-turbo few-shot classification and Pydantic-validated accept, reject, and retry routing to filter approximately 70% of low-value content, then cleaned, deduplicated, chunked, enriched, and preserved source provenance across a 50K+ chunk corpus through incremental Parquet updates completed in under 30 minutes.
- Improved relevant-context retrieval from 42% to 87% on approximately 50 manually labeled queries by combining 384-dimensional MiniLM SentenceTransformer embeddings, FAISS HNSW candidate search, lexical retrieval, weighted Reciprocal Rank Fusion, and cross-encoder reranking.
- Engineered a bounded LangGraph/LangChain RAG agent for intent classification, canonical entity and conversational-context resolution, and multi-intent decomposition; constrained execution through Pydantic-validated state, fixed retrieval, reranking, and verification tools, per-tool timeouts, and a five-action budget with evidence-gated generation and explicit clarification, retry, and abstention paths.
- Built a reinforcement-learning feedback loop by training a DistilBERT reward model on answer-level user feedback and using trIX PPO to optimize the ranking policy against the learned reward signal, allowing validated user preferences to influence subsequent retrieval rankings.
- Deployed SageMaker-hosted retrieval and reranking services with a Streamlit decision-support application through CI/CD using automated tests, versioned artifacts, and rollback; instrumented LangSmith and CloudWatch tracing to analyze retrieval failures, tool execution, grounding, latency, and agent behavior.

Accenture

February 2021 - July 2023

Packaged App Development Analyst

- Partnered with quantitative analysts and business stakeholders to translate investment-research needs into requirements for an enterprise financial-analytics platform; built PySpark and SQL pipelines that transformed point-in-time fundamentals across thousands of public companies into partitioned Parquet datasets on Amazon S3.
- Developed pooled LSTM/XGBoost and per-company SARIMA models to forecast company KPIs up to two years forward, achieving 92% directional accuracy during stable market regimes and improving transition-period performance by 35% over the existing forecasting baseline.
- Fine-tuned FinBERT on earnings calls, market news, and analyst reports using forward price movement as the supervision signal; integrated the resulting market-impact score with structured financial features to help institutional analysts identify company-growth, valuation, and competitor risks not captured by quarterly fundamentals.

PROJECT

PyLinguist: Cross-Lingual Python Translation

November 2024 – Present

Research project | Submitted to ACL BEA

- Built a two-stage Python translation system across seven languages using token-aware translation, a curated dictionary of 234 Python terms, and Google Translate followed by GPT-4o or Claude 3.5 Haiku refinement, preserving code syntax and structure across diverse writing systems.
- Developed a back-translation evaluation framework combining BLEU, syntax validity, structural preservation, token matching, semantic similarity, and bilingual human review; across 10 held-out programs per language, GPT-4o achieved a 92.5% average overall score versus 61.5% for the baseline and maintained 100% syntax validity, while identifying that additional few-shot examples degraded Claude's performance in some languages without affecting GPT-4o's.

TECHNICAL SKILLS

Languages & Data: Python, SQL, C, PySpark, ETL, Parquet

GenAI & Agents: RAG, LangGraph, LangChain, Pydantic, Prompt Engineering, Evidence-Gated Generation

Retrieval & Preference Learning: SentenceTransformers, FAISS, HNSW, Hybrid Search, Weighted RRF, Cross-Encoder Reranking, DistilBERT, PPO, trIX

ML & Evaluation: PyTorch, XGBoost, LSTM, SARIMA, FinBERT, Deep Learning, Time-Series Forecasting, probability, statistics, data distributions, LLM/Agent Evaluation, Model Validation, BLEU

Cloud & MLOps: AWS (SageMaker, Lambda, S3, SQS, Glue, Athena, CloudWatch), Azure, MLflow, LangSmith, CI/CD, Automated Testing, Streamlit

EDUCATION

George Mason University

August 2023 - May 2025

M.S. in Computer Science, Machine Learning Concentration | GPA: 3.90/4.00

Honors: Distinguished Academic Achievement Award

CERTIFICATIONS

- AWS Certified Machine Learning Engineer - Associate (MLA-C01)
- AWS Certified Developer - Associate (DVA-C02)